AGENT KIO

Project Plan & Team Responsibilities

Customer Message Routing | 3 Students | Presentation: Monday, 26 October 2026

1. Project Goal

Build a small intent router that reads a customer message and chooses the correct destination: Support (flag 0) or Sales (flag 1). Compare two approaches: a traditional machine-learning classifier trained on examples and a decision model/API that follows routing instructions. This is a routing experiment—not a complete chatbot.

2. Required System Flow

Customer message → Intent Router → Flag (0 or 1) → Simulated Support or Sales function

Support handles problems, complaints, and existing requests. Sales handles new purchases, service packages, and quotes. For mixed-intent messages, prioritize an existing problem and route to Support.

3. Dataset and Evaluation Rules

- Create exactly 60 synthetic English messages: 30 Support and 30 Sales.
- Agree on labels before testing. Include varied wording, paraphrases, negation, and mixed-intent examples.
- Split into 40 training messages and 20 test messages, balanced by class: 20/20 for training and 10/10 for testing.
- Keep near-duplicate messages in the same split to reduce data leakage.
- Evaluate both approaches on the same 20 test messages. Report actual correct/incorrect counts as well as accuracy; each test error changes accuracy by 5 percentage points.

4. Divide the Work Between Three Students

Team member	Main responsibility	Deliverables
Student 1 **Problem & Dataset**	Explain the business problem and architecture. Create the 60 messages, define consistent labels, make the balanced 40/20 split, and show sample rows.	Dataset table/file; label rules; architecture diagram; slides on problem and dataset.
Student 2 **Decision Model**	Choose a decision model/API or suitable open-source alternative. Write clear routing instructions, validate its output, test on the 20 test messages, and analyze errors. Document model/license/setup.	Working decision-router code; outputs and test results; sources and limitations.
Student 3 **Traditional ML & Comparison**	Build the traditional ML classifier using an allowed option from the brief. Explain numeric text features and settings, test it on the same 20 messages, calculate metrics, and compare both approaches.	Working ML code; metrics/confusion matrix; comparison table; recommendation.

5. Shared Tasks (All Three Students)

- Agree on routing rules and labels together before evaluating either approach.
- Integrate both approaches into one consistent demo. Map Support to 0 and Sales to 1, then call a simple function that prints the selected agent.

- Check that both approaches use the identical test messages and that no test examples were used to train or tune the models.
- Review each other's results, verify all claims, and rehearse the presentation with every student speaking.

6. Suggested Implementation Order

1. **1. Define —** Confirm labels, mixed-intent rule, and model choice.
2. **2. Prepare —** Create and inspect the 60-row dataset; split it 40/20.
3. **3. Build —** Student 2 builds the decision-model route; Student 3 builds the ML route.
4. **4. Evaluate —** Run both on the same 20 test messages and record errors and metrics.
5. **5. Integrate —** Connect each predicted flag to the simulated Support/Sales functions.
6. **6. Present —** Show architecture, sample data, outputs, results, limitations, and recommendation.

7. Important Notes

- The brief allows a linear classifier, Logistic Regression with degree-2 polynomial features, or an SVM with a linear or polynomial kernel. Confirm the exact implementation with your instructor if unsure. TF-IDF is a possible text-to-number feature method, not explicitly mandated in the brief.
- The decision model can be Jev or another suitable decision/classification model. Document its license, setup, and output validation.
- Do not claim that 60 messages prove real-world reliability. With only 20 test messages, each error changes accuracy by 5 percentage points. Present this as a small comparative experiment.
- A full chatbot, real LLM-based specialist agents, search tools, and RAG pipeline are outside this project's first-stage scope. Simulate agents with functions that print their names.
- The presentation is limited to 8 minutes and all three students must speak. The brief says no separate submission files are required; this document is your team's planning guide.